

¹Muna Hameed Khalaf,²Behrouz Tousi

Fault Location and Detection in Power Distribution Systems with Presence of Distributed Generations using Improved CNNs



Abstract:

This paper introduces a deep learning approach for addressing fault detection and location issues in power distribution grids. The proposed method utilizes a 20-layer deep neural network (CNN) that employs the ReLU activation function in the hidden layers and the Softmax function in the pre-classification layer. The network's input layer dimensions are 224 x 224 pixels, with each input image being a composite of seven images, each sized 32 x 224 pixels. Data is gathered using continuous wavelet transform and Hilbert transform on each signal, followed by feature extraction and conversion into RGB images. The signals include three-phase voltages, currents, and zero-sequence voltages, collected across different locations, times, and fault resistances for use in the training and testing of the deep neural network. The output layer categorizes faults into ten classes, covering types such as single-phase to ground, two-phase, and three-phase faults. Additionally, the location identification layer categorizes faults into five classes corresponding to five zones on a specific feeder. The proposed method is validated using a test power system simulated in MATLAB 2020b, which includes six main feeders and two sub-feeders. The validation is conducted under two scenarios: one without distributed generation sources and with data noise at SNR = 10dB, and another with distributed generation sources and SNR = 40dB. The method demonstrated strong performance in all cases, and the trained network successfully detected and located faults in previously untested areas.

Keywords: fault location, power distribution systems, deep learning, distributed generation

1- INTRODUCTION

In the field of fault location and detection in power distribution systems with the presence of distributed generation, numerous studies have explored various techniques and approaches. For instance, in [1], the authors discussed fault location and detection methods specifically designed for power distribution systems integrated with distributed generation, highlighting the unique challenges and solutions in this context. In another study [2], a hybrid method combining k-means and VSS LMS learning for RBF networks was proposed for short-term load forecasting, which can also be relevant in fault detection scenarios by providing accurate system state predictions. The application of deep learning methods in electrical systems has been extensively reviewed in [3], where the authors emphasized the growing importance of advanced neural networks like CNNs in improving fault detection accuracy. In [4], the use of a deep learning-based approach leveraging the Hilbert-Huang Transform and CNNs for fault classification in power distribution systems was demonstrated, showcasing the effectiveness of such techniques in enhancing fault classification precision. Similarly, [5] explored the use of continuous wavelet transform combined with CNNs for earth fault detection in resonant grounding distribution systems, providing evidence of the method's robustness under various fault conditions. The authors in [6] applied machine learning techniques to locate faults in distribution systems with distributed generation, offering a comparative analysis of different methods in such scenarios. Additionally, the research presented in [7] investigated artificial intelligence techniques for fault location and detection in distributed generation power systems, emphasizing the role of AI in addressing the complexities introduced by distributed energy resources. The study in [8] introduced a novel

¹ 1,2 Department of electrical and computer engineering, Urmia University, Urmia, Iran.

m.hameedkhalaf@urmia.ac.ir, b.tousi@urmia.ac.ir

Corresponding author: Behrouz Tousi, b.tousi@urmia.ac.ir

approach using deep graph convolutional networks for fault location in power distribution systems, further advancing the field by integrating graph-based learning methods. Artificial neural networks were also utilized for fault section identification and location in distribution networks, as detailed in [9], providing insights into the practical applications of ANNs in real-world systems. A comparative review of fault location methods in distribution networks was conducted in [10], offering a broad perspective on the existing techniques and their relative strengths. In [11], the authors focused on non-linear high impedance fault distance estimation using a continually online-trained neural network, demonstrating the advantages of adaptive learning in fault detection. The study in [12] explored the use of traveling waves and ANNs for fault location in power systems, contributing to the development of more accurate fault location methods. Single-phase fault location in distribution systems with distributed generation was addressed in [13], where the authors proposed a scheme based on ANNs, highlighting the adaptability of neural networks in diverse fault scenarios. In [14], a machine learning-based approach for designing a power distribution network fault data collector was presented, aiming to improve fault detection, location, and classification capabilities. The research in [15] focused on fault location in unbalanced power distribution feeders with distributed generation using neural networks, adding to the growing body of work on ANN-based fault location methods. The study in [16] provides an in-depth analysis of MATLAB-based deep learning methods, emphasizing their applicability in neural networks and artificial intelligence, which are crucial for fault detection in power systems. The work in [17] explores the use of continuous wavelet transform for extracting signal frequency information, a technique that plays a pivotal role in enhancing the accuracy of fault detection when integrated with deep learning models. Further advancements in the use of deep learning for fault detection are demonstrated in [18], where the authors employed LSTM networks for fault detection, diagnosis, and location in large-scale multi-machine power systems. This approach highlights the potential of recurrent neural networks in handling the sequential nature of fault signals. In [19], a comparison framework for distribution system outage and fault location methods was presented, providing valuable insights into the effectiveness of various approaches, particularly in systems with distributed generation. The application of deep learning in fault location identification and type classification in active distribution grids was further investigated in [20], where the authors demonstrated the utility of deep learning-based models in enhancing fault management in modern power systems. The research in [21] presented a deep learning-based relay for online fault detection, classification, and location in grid-connected microgrids, underscoring the importance of real-time fault management in maintaining grid stability. In [22], extreme learning machines were utilized for fast and accurate fault detection and classification in transmission lines, showcasing the speed and efficiency of machine learning models in fault management. The study in [23] applied deep residual networks for fault detection in wind turbines, illustrating the versatility of deep learning models across different types of power systems. Moreover, [24] proposed a CNN-based transformer model for fault detection in power system networks, further demonstrating the effectiveness of convolutional neural networks in fault detection tasks. Challenges and opportunities in the application of deep learning for process fault detection and diagnosis were reviewed in [25], providing a comprehensive overview of the field and identifying key areas for future research. A high-speed fault detection and localization scheme for low-voltage DC microgrids was introduced in [26], highlighting the importance of rapid fault response in maintaining system reliability. The work in [27] discussed a protection scheme for microgrids dominated by solid-state converters, focusing on the harmonic content-based detection method, which is crucial for accurately identifying faults in such systems. In [28], an artificial neural network-based approach was used for fault detection and classification in electrical power transmission systems, reinforcing the applicability of ANNs in power system fault management. Sequential overlapping differential transform was employed for microgrid fault detection in [29], offering an innovative method for improving fault detection accuracy. A fault detection scheme for multi-terminal direct current (MTDC) systems using a superconducting fault current limiter was proposed in [30], demonstrating the integration of advanced fault detection techniques in complex power networks. In [31], an open circuit fault detection method for H-bridge converters using transistor logic fault detection modules was introduced, contributing to the development of more reliable power electronics systems. The detection of line-to-ground and line-to-line faults based on fault voltage analysis in photovoltaic (PV) systems was explored in [32], emphasizing the need for specialized fault detection methods in renewable energy systems. Power hardware-in-the-loop testing to analyze fault behavior of smart inverters in distribution networks was presented in [33], offering a practical approach to evaluating and improving fault detection in modern power systems. The research in [34] applied LSTM networks for power system load forecasting, which is closely related to fault detection as accurate

load forecasting can prevent potential faults. Quick event detection in power systems using LSTM networks was discussed in [35], further highlighting the role of deep learning in enhancing the responsiveness of fault detection systems. The application of LSTM-based models for regional integrated energy systems was explored in [36], providing insights into the use of deep learning for managing complex energy networks. Short-term load forecasting and associated weather variables prediction using ResNet-LSTM-based deep learning was investigated in [37], showcasing the application of hybrid deep learning models in power systems. An improved LSTM model was used for fault diagnosis of modular multilevel converters in [38], demonstrating the potential of deep learning in handling complex fault scenarios. The study in [39] focused on simulation and state prediction of nuclear power systems using LSTM neural networks, indicating the broad applicability of deep learning across different power sectors. Electricity load forecasting in Thailand using deep learning models was discussed in [40], further emphasizing the global relevance of deep learning in power systems management. Finally, the application of machine learning methods in fault detection and classification of power transmission lines was extensively reviewed in [41], providing a comprehensive survey of the state-of-the-art techniques and their practical implications in enhancing the reliability and resilience of power distribution systems, particularly in the presence of distributed generation.

2- PROBLEM STATEMENT

2.1 Definition of Fault

A fault in a distribution system is an unauthorized deviation from its standard operating conditions. Faults can be classified by their duration: some faults exist for a short period and then revert to normal conditions, referred to as transient faults. Permanent faults remain until the short circuit is identified and resolved. If transient faults are not cleared, they may convert into permanent faults [1].

2.2 Symmetrical and Asymmetrical Faults

There are two types of faults that distribution systems typically encounter: balanced faults and unbalanced faults, which are also referred to as symmetrical and asymmetrical faults, respectively [1]. The power transfer capacity of a transmission line is reduced to zero during a symmetrical short circuit, whereas in an asymmetrical short circuit, a portion of the previous power continues to be transmitted. In addition to the reduction in power transfer capacity, short circuit currents can damage system devices and equipment; therefore, short-circuited areas must be promptly isolated from the system [2].

2.3 Series and Shunt Faults [1]

Series faults are categorized based on frequency, with an increase in voltage and a decrease in current in the affected phases. The key characteristic of shunt faults is the increase in current along with a drop in voltage and frequency.

2.4 Causes of Faults [2]

- Lightning
- Failure of equipment and system components
- Environmental conditions such as strong winds, severe icing, etc.
- Collisions of ground vehicles with towers and aerial vehicles with transmission line conductors
- Bird collisions with transmission line conductors or the intrusion of animals into substations and switchgear
- Falling trees onto transmission line conductors
- Random factors and unforeseen events

2.5 Main Categories of Faults in Distribution Systems [1]

- Single Line to Ground Fault (SLGF)
- Line to Line Fault (LLF)
- Double Line to Ground Fault (DLGF)
- Three-phase to Ground Fault (LLLGF)

2.5.1 Single Line to Ground Fault (SLGF)

A Single Line to Ground Fault occurs when one of the three-phase conductors in a distribution system makes contact with the ground. The causes of such faults include wind, animal collisions, or the line falling to the ground. Approximately 70% of faults occurring in power distribution systems are of this type. Types of Single Line to Ground Faults are illustrated in Fig. 1, where R_f represents the fault resistance.

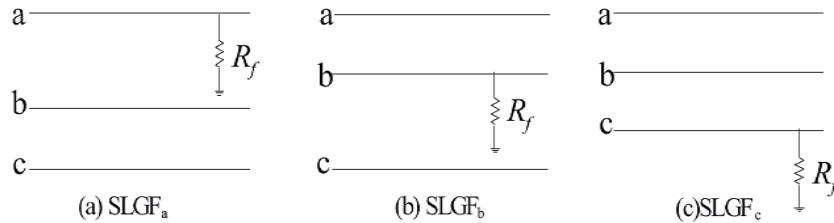


Fig.1 Types of Single Line to Ground Fault [1]

2.5.2 Line-to-Line Fault (LLF)

A line-to-line fault occurs when severe wind causes one phase to make contact with another phase. This type of fault can occur in both aerial and underground transmission systems. Approximately 15% of faults in power distribution systems are due to line-to-line faults. Fig. 2 illustrates the various types of line-to-line faults.

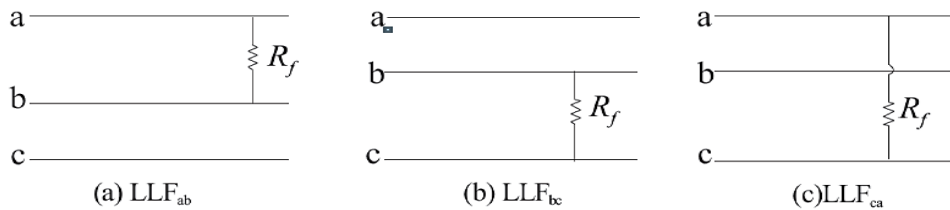


Fig. 2 Types of Line-to-Line Faults (Phase-to-Phase) [1]

2.5.3 Double Line to Ground Fault (DLGF)

In a double Line to Ground Fault, two phases are involved instead of one, as in the case of a single-phase to ground fault. This type of fault typically occurs when a tree falls and connects two phases to each other and to the ground.

This type of fault is a serious event that leads to significant asymmetry, and if not addressed in a timely manner, it may escalate to a three-phase fault.

Approximately 10% of faults are classified as double Line to Ground Fault.

Fig. 3 illustrates the various types of double Line to Ground Faults:

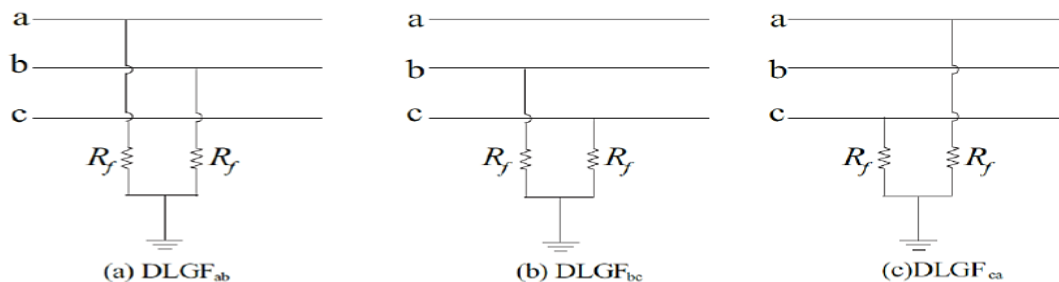


Fig. 3 Types of Double Line to Ground Faults [1]

2.5.4 Three-Phase to Ground Fault (LLLGF)

A three-phase to ground fault is also known as a symmetrical fault. It may occur due to the failure of a piece of equipment, the collapse of a power pole, or a connection between a line and the other phases.

In practical scenarios, three-phase to ground faults are often rare, representing only about 5% of all faults; however, they are considered the most dangerous type of fault. In a three-phase to ground fault, the line-to-ground voltage is zero, and the fault current is significantly large. Fig. 4 illustrates a three-phase to ground fault:

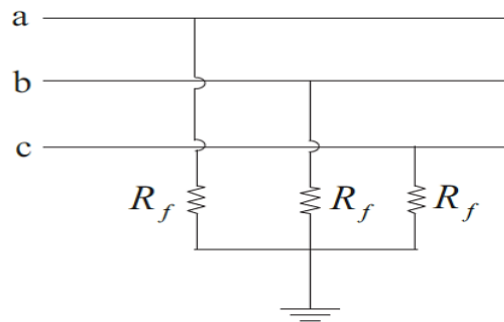


Fig. 4 Three-Phase to Ground Fault [1]

The occurrence percentage and severity of various fault types are presented in Table 1:

Table 1 Percentage Occurrence of Fault Types [1]

Fault	Occurrence Percentage (%)	Severity
SLGF	70	Less Severity
LLF	15	Less Severity
DLGF	10	Less Severity
LLLGF	5	More Severity

Considering Table 1, it is evident that most faults in the distribution system are of the single-phase to ground type. Nevertheless, to prevent damage to the network and consumers, all types of faults must be taken into account.

The percentage of fault occurrence resulting from power system elements is shown in Table 2:

Table 2: Fault occurrence caused by power system elements [1].

Elements of the power system	Occurrence Percentage (%)
Transformers	10
Airlines	50
Underground cables	9
Switchgear	12
CT, RT relays, control equipment and...	12
Generators	7

Given the various types of faults in distribution systems and the extent of damage they can cause to network equipment, transmission lines, cables, and other network elements, as well as the critical impact on system reliability, addressing fault detection becomes particularly significant. Even faults with lower occurrence rates should not be overlooked. For example, while the occurrence rate of a three-phase to ground fault is only 5%, it is one of the most dangerous fault types, as it can result in very high fault currents leading to potentially irreversible human and economic damages. Therefore, in evaluating different fault detection schemes, it is essential to consider all types of faults.

3- PROPOSED METHODS

The architecture of the CNN for fault detection and classification is illustrated in Fig. 5. The image that serves as the network input, shown in the Fig., is generated after feature extraction from signals during faults in RGB format. This image is then processed through the defined layers of the network, resulting in the generation of additional images with varying features based on the number of filters in each layer and their coefficients. Between each convolutional layer, a ReLU activation layer is present, while a dropout layer is placed before the fully connected layer to prevent overfitting during the training process. The softmax layer before the classification layer is used for normalization; the same procedure applies to localization, with the only difference being the number of classes in the final layer.

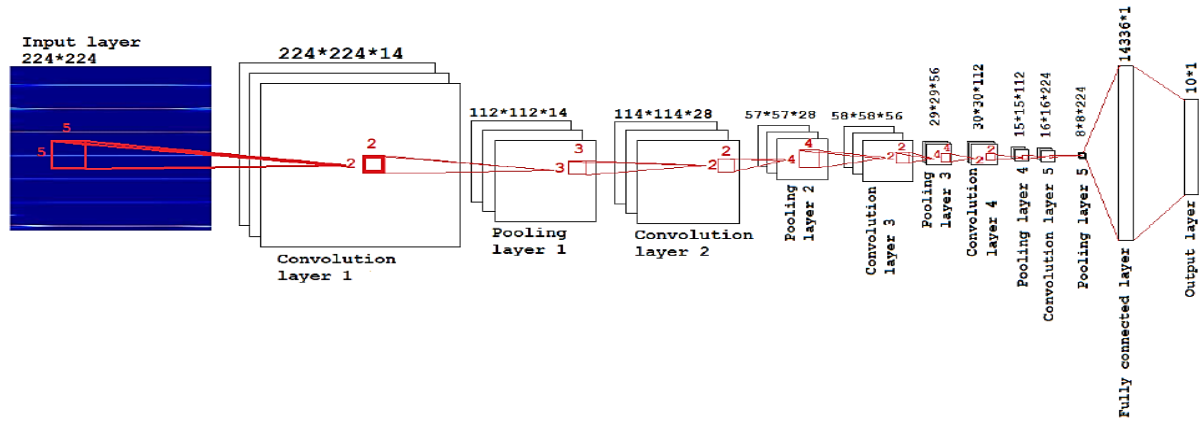


Fig. 5: The structure of the CNN for fault detection and classification.

The details of the deep neural network architectures for CNNs addressing both fault detection and localization are presented similarly in Table 3.

Table 3: The details of the CNN architectures for the paper.

Activator function	padding	Stride	Filter size	Output size	Type	Layer
ReLU	2	1	5*5	224*224*14	Convolution	C1
ReLU	---	2	2*2	112*112*14	Pooling	P1
ReLU	2	1	3*3	114*114*28	Convolution	C2
ReLU	---	2	2*2	57*57*28	Pooling	P2
ReLU	2	1	4*4	58*58*56	Convolution	C3
ReLU	---	2	2*2	29*29*56	Pooling	P3
ReLU	2	1	4*4	30*30*112	Convolution	C4
ReLU	---	2	2*2	15*15*112	Pooling	P4
ReLU	2	1	4*4	16*16*224	Convolution	C5
ReLU	---	2	2*2	8*8*224	Pooling	P5
Softmax	---	---	---	14336*1	Fully connected	F1

The block diagram for error detection and classification using continuous wavelet transform and Hilbert transform is presented in Fig. 6 of this paper, depicting the steps involved in these processes sequentially.

In both the classification and localization tasks, 70% of the data was used for training and 30% for validation.

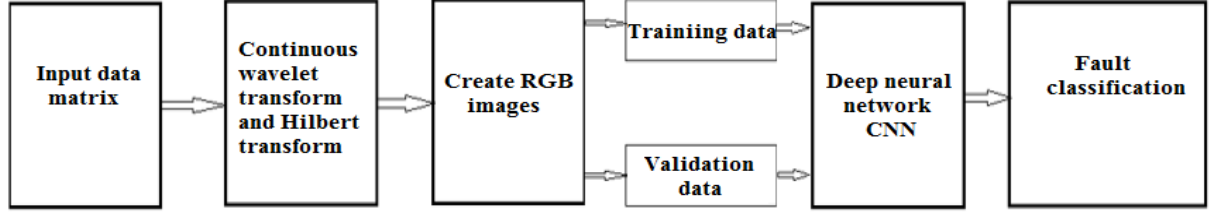


Fig. 6: The block diagram for fault classification.

The block diagram for error localization using continuous wavelet transform and Hilbert transform is illustrated in Fig. 7, depicting the sequential execution of these operations.

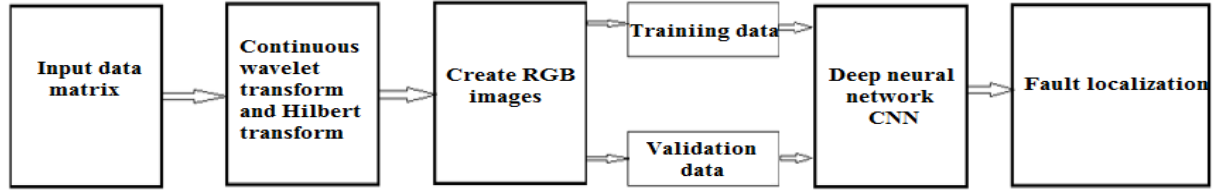


Fig. 7: The block diagram for fault localization.

The structure of the CNN (Convolutional Neural Network) for error localization is presented in Fig. 8. The difference between this network and the error detection network lies solely in the last layer, which is determined by the number of zones selected on a chosen feeder.

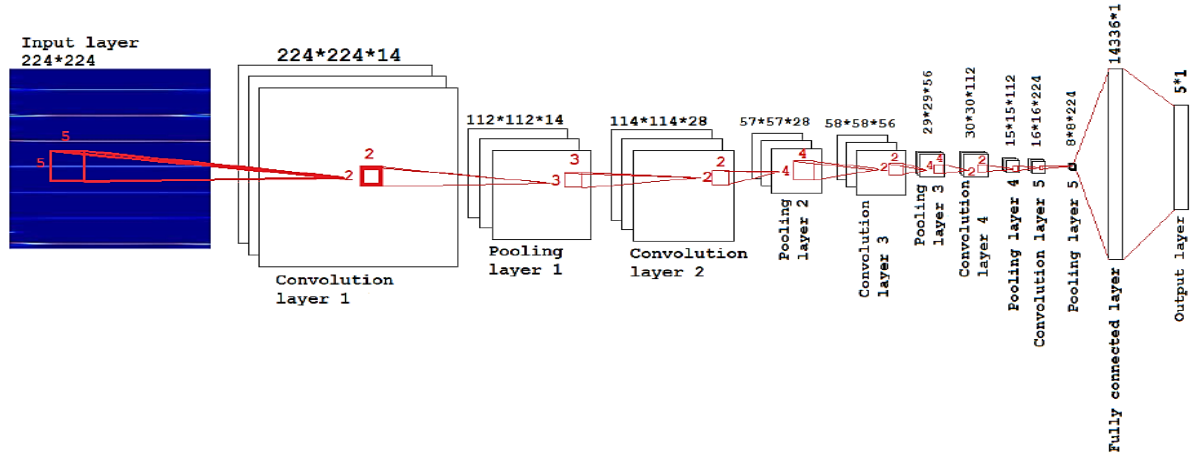


Fig. 8: The structure of the CNN (Convolutional Neural Network) for error localization.

3-7-1 Proposed CNN Structure in the Simulation Process

After generating images for each fault class as previously described and placing them in their respective class folders, we randomly select seventy percent of these images for training and thirty percent for validation of the designed deep convolutional neural network aimed at addressing the fault type detection problem.

The first layer, or the input layer for the image, is designed based on the fact that the obtained images are stored in RGB format after feature extraction. Consequently, each image consists of three channels:

- Red channel
- Green channel
- Blue channel

Thus, the first step is to ensure that the dimensions of the input image layer match the dimensions of the obtained images, which should be 224 by 224 pixels and contain three channels, allowing for proper image reading.

```
imageInputLayer ([224 224 3], 'Name', 'input')
```

The second layer is a two-dimensional convolution layer with fourteen filters, configured with a padding of 2 and a stride of 1. Due to the presence of padding 2, this layer transforms each 224x224 image into fourteen new images, each also sized at 224x224.

```
convolution2dLayer(5,14,'Stride',1,'Padding',2,'Name','conv1')
```

The third layer is an activation function; up to the second layer, we only performed convolution operations. Typically, an activation function is applied after the convolution layer in CNN architectures. We utilized the ReLU activation function, which, as explained in Chapter 3, outputs zero for negative values and retains the original value for inputs greater than zero.

```
reluLayer('Name','relu1')
```

The fourth layer is a two-dimensional max pooling layer. The pooling operation in this layer has a size of 2x2 with a stride of 2, meaning it moves across the image produced by the previous layer by two pixels at a time. Since we are using max pooling, it extracts the maximum value from each 2x2 section of the image. Consequently, the dimensions of the new image obtained are half of those of the previous layer, resulting in a size of 112 by 112 pixels.

```
maxPooling2dLayer(2,'Stride',2,'Name','pool1')
```

The fifth layer is a two-dimensional convolution layer with twenty-eight filters. This layer transforms the images from the previous layer to dimensions of 114 by 114 pixels, as specified by the command

```
convolution2dLayer(3,28,'Stride',1,'Padding',2,'Name','conv2')
```

The sixth layer is the second activation function of the deep neural network. As mentioned, it is positioned following the convolution layer

```
reluLayer('Name','relu2')
```

The seventh layer is a two-dimensional max pooling layer, which reduces the dimensions of the images from the previous layer to 57 by 57 pixels, as specified by the command:

```
maxPooling2dLayer (2, 'Stride', 2, 'Name', 'pool2')
```

The eighth layer is a two-dimensional convolution layer with fifty-six filters. This layer transforms the images from the previous layer to dimensions of 58 by 58 pixels, as specified by:

```
convolution2dLayer (4, 56, 'Stride', 1, 'Padding', 2, 'Name', 'conv3')
```

The ninth layer is the third activation function of the deep neural network, positioned following the convolution layer, as previously mentioned:

```
reluLayer ('Name', 'relu3')
```

The tenth layer is another two-dimensional max pooling layer, which reduces the dimensions of the images from the previous layer to 29 by 29 pixels

```
maxPooling2dLayer (2, 'Stride', 2, 'Name', 'pool3')
```

The eleventh layer is a two-dimensional convolution layer with one hundred twelve filters. This layer converts the images from the previous layer to dimensions of 30 by 30 pixels, as specified by:

```
convolution2dLayer (4, 112, 'Stride', 1, 'Padding', 2, 'Name', 'conv4')
```

The twelfth layer is the fourth activation function of the deep neural network, also positioned following the convolution layer

```
reluLayer ('Name', 'relu4')
```

The thirteenth layer is yet another two-dimensional max pooling layer, which reduces the dimensions of the images from the previous layer to 15 by 15 pixels

```
maxPooling2dLayer (2, 'Stride', 2, 'Name', 'pool4')
```

The fourteenth layer is a two-dimensional convolution layer with two hundred twenty-four filters. This layer transforms the images from the previous layer to dimensions of 16 by 16 pixels, as specified by the command:

```
convolution2dLayer (4, 224, 'Stride', 1, 'Padding', 2, 'Name', 'conv5')
```


The fifteenth layer is the fifth activation function of the deep neural network, positioned following the convolution layer, as previously mentioned

`reluLayer ('Name', 'relu5')`

The sixteenth layer is a two-dimensional max pooling layer, which reduces the dimensions of the images from the previous layer to 8 by 8 pixels.

`maxPooling2dLayer (2, 'Stride', 2, 'Name', 'pool5')`

The seventeenth layer is a dropout layer. As explained in Chapter Three, this layer is crucial for preventing overfitting in deep neural networks. In this layer's settings, we have set it to 70%, meaning that in each training iteration, 70% of the neurons are randomly deactivated

`dropout Layer (0.7, 'Name', 'drop')`

The eighteenth layer is a fully connected layer. This layer converts two-dimensional feature maps into a one-dimensional feature vector for further processing in feature representation. The fully connected layer allows us to present the network's output as a vector of a specified size. Neurons in a fully connected layer are connected to all neurons in the previous layer, similar to what is observed in conventional neural networks. Therefore, matrix multiplication followed by the addition of a bias term can be used to compute the outputs of all neurons (activations) at once. In this layer, we specify the number of classes:

`fullyConnectedLayer (10, 'Name', 'fc1')`

The nineteenth layer is the softmax layer, which must be applied before the classification layer for the purpose of normalization

`softmaxLayer ('Name', 'sf')`

The twentieth layer is the classification layer. This layer serves as the output of the network and infers the number of classes based on the size of the previous layer

`classificationLayer ('Name', 'output')`

3-7-2 Learning Parameters for the Training Process

- The Adam optimization method, one of the most popular learning algorithms, is employed.
- A mini-batch size of 32 is used, meaning that at each iteration, the network selects 32 images.
- The number of epochs is set to 42.
- The initial learning rate is set to 1e-4
- To reduce training time and enhance the training process, a GPU is utilized.

`Options = training Options ('adam', ...`

`'MiniBatchSize', 32, ...`

`'MaxEpochs', 42, ...`

`'InitialLearnRate', 1e-4, ...`

`'ValidationData', imgsValidation, ...`

`'ValidationFrequency', 100, ...`

`'ValidationPatience', Inf, ...`

`'Verbose', 1, ...`

`'Execution Environment', 'gpu', ...`

`'Plots', 'training-progress');`

Considering that localization not only involves error type detection but also the determination of the location of the error occurrence, and due to the similarity of the neural network used for localization, we will not elaborate further on its specific details.

The system and software version used for simulating and training the deep neural network are as follows:

- Processor: Core™ i7-10510u 2.3-GHz IntelCore™
- Operating System: Win64

- MATLAB Version: 2020b
- RAM: 8.00 GB

4- RESULTS AND ANALYSIS

4.1 Study System

As we know, the most challenging scenario for fault detection and localization in distribution systems occurs when there are distributed generation (DG) sources within the network. In this context, we have analyzed the system depicted in Fig. 4-1 for the active power distribution system. The feeders are represented by overhead Feeder, labeled as O_L , and Cable Feeder, labeled as C_L , while points F1 to F14 indicate the locations of potential faults.

DG1 and DG2 are distributed generation sources connected to feeders 4 and 5 at the points indicated in Fig. 9 to the distribution system. The impedance of the DGs is $(3.64 + j21.7) \Omega$, and their active and reactive power capacities are $(0.25 + j0.15)$ MVA, respectively. The study system is also represented in a single-line diagram without distributed generation sources in Fig. 10.

Main Transformer Specifications:

- Voltage: 110 kV/10 kV
- Power: 31.5 MVA

Load Specifications for Each Feeder:

- $L = 0.5 + j0.25$ MVA

Table 4 presents the specifications for both aerial and cable lines, including zero and positive sequence parameters, as well as resistance, inductance, and capacitance values. The parameters for the power distribution system remain unchanged in both scenarios, whether or not distributed generation sources are present.

Table 4: Parameters Related to the Distribution System Lines

Parameter	Unit	Line type	
		O_L	C_L
R_0	Ω / km	0.23	2.70
R_1	Ω / km	0.17	0.27
L_0	mH / km	5.48	1.02
L_1	mH / km	1.21	0.26
C_0	$\mu\text{F} / \text{km}$	0.01	0.28
C_1	$\mu\text{F} / \text{km}$	0.01	0.34

The single-line diagram of the active power distribution system is shown in Fig. 9.

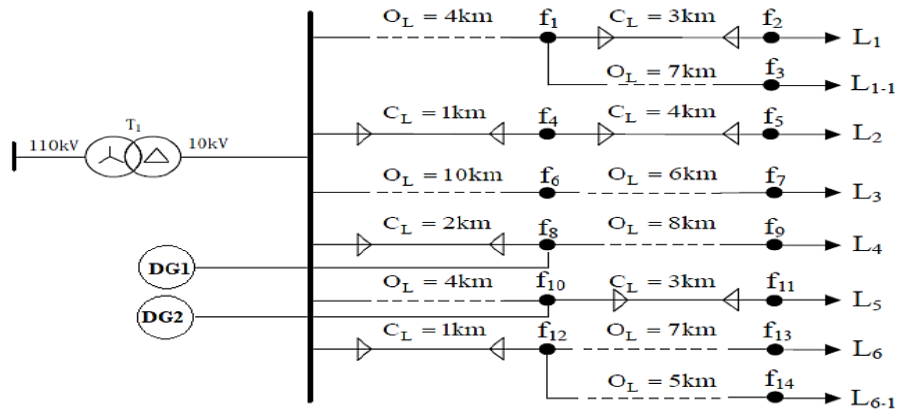


Fig. 9: Single-line Diagram of the Active Power Distribution System

The single-line diagram of the power distribution system without the presence of distributed generation sources is shown in Fig. 4-2.

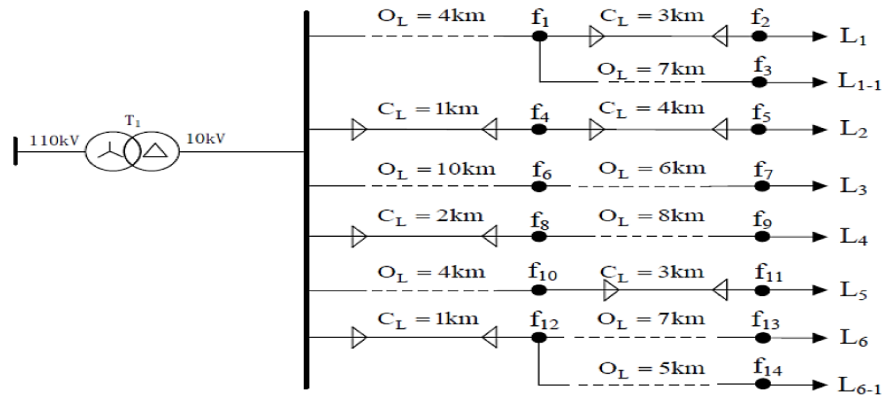


Fig. 10: The single-line diagram of the distribution system without the presence of distributed generation sources.

In both Figs. 9 and 10, the parameters for the lines, transformers, type of lines, and length of each feeder are identical; the only difference between them is the presence of distributed generation sources in the system depicted in Fig. 9.

The active power system under study is illustrated in Fig. 11, which has been simulated in the Simulink environment of MATLAB.

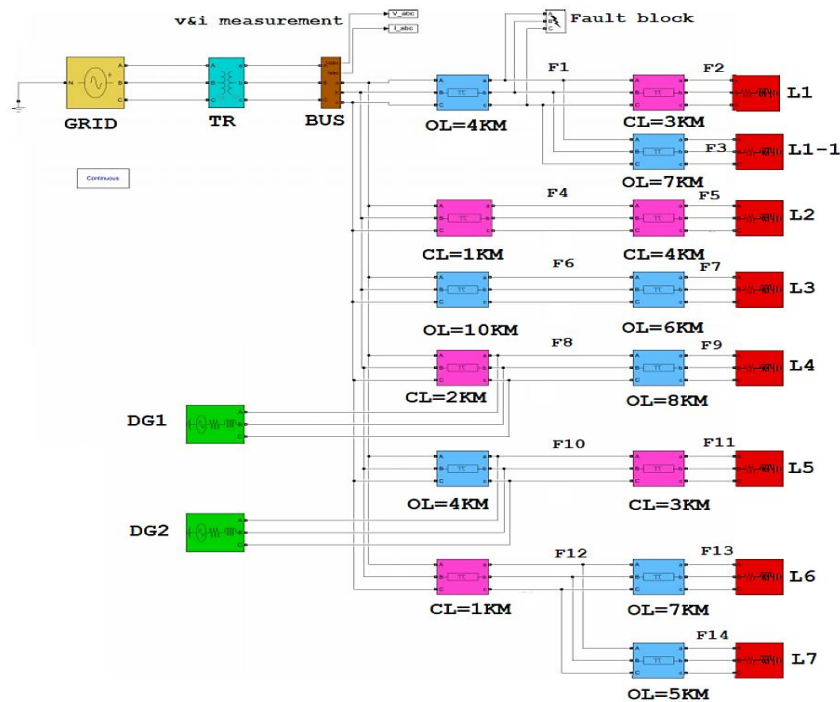


Fig. 11: The active power distribution system under study.

The power system without the presence of distributed generation sources is illustrated in Fig. 12, which has been simulated in the Simulink environment of MATLAB.

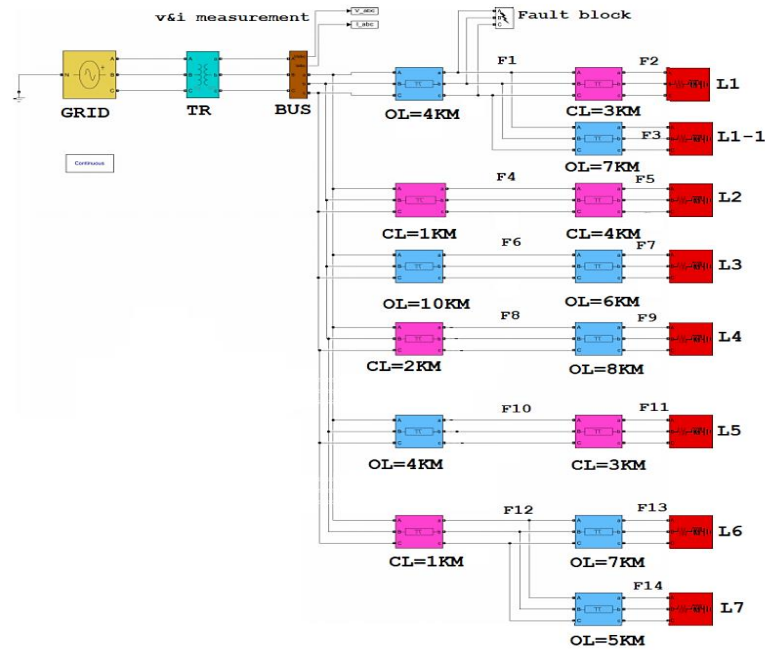


Fig. 12: Active Power Distribution System Under Study Without Distributed Generation Sources.

4.2 Simulation of Faults at Designated Points for Data Collection

To train a deep neural network, we need data that learns the conditions during the fault to predict it in the future.

In this paper, after applying various faults at different locations and angles on the studied distribution system—which will be presented in a table in subsequent sections—we collect data. For each type of fault, we store three voltage signals and three current signals with a sampling time of 0.001 seconds for a duration of 0.2 seconds, coinciding exactly with the fault duration, in the form of a matrix. This chapter thoroughly explains how to construct the data matrix for training the deep neural network.

Additionally, for each type of fault, we include the zero-sequence voltage signal. In essence, for one class of fault at a specific location and angle, we generate a matrix with 7 rows and 2000 columns; the first three rows contain the three-phase voltage values, the next three rows contain the three-phase current values, and the final row consists of the zero-sequence voltage. The structure of this matrix is represented in equation (1):

$$\text{Fault matrix} = [V_a \ V_b \ V_c \ I_a \ I_b \ I_c \ V_0]' \quad (1)$$

Table (5) illustrates the data sampling for the case without distributed generation sources.

Table 5: Fault Table for the Case Without the Presence of Generation Sources.

Network	Fault types	Fault locations	Fault resistance (ohm)	Fault Inception Angle (FIA) (degree)	Number
Fig 4-4	AG	f1, f2, ..., f14	0.001	0,45,90	126
	BG				
	CG				
Fig 4-4	AB	f1, f2, ..., f14	0.001	0,45,90	126
	AC				
	BC				
Fig 4-4	ABG	f1, f2, ..., f14	0.001	0,45,90	126
	ACG				

	BCG				
Fig 4-4	ABCG	f1, f2, ..., f14	0.001	0,45,90	42
Total					420

Based on the table above, for each fault class at a single location, observed from three angles and with one fault resistance, the resulting matrix consists of 21 rows ($3 * 7 = 21$) and 2000 columns. This matrix includes seven rows representing three-phase voltage and current signals, along with one zero-sequence voltage signal. Consequently, for 14 locations and three angles with one fault resistance for a single fault class, the resulting matrix contains 294 rows and 2000 columns. The overall data matrix for various fault classes comprises 2940 rows and 2000 columns. Table 6 provides a comprehensive overview of the number and types of faults occurring at different locations, angles, and fault resistances within an active distribution system.

Table 6: Fault Table in an Active Distribution System.

Network	Fault types	Fault locations	Fault resistance (ohm)	Fault Inception Angle (FIA) (degree)	Number
Fig 4-3	AG BG CG	f1, f2, ..., f14	0	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	672
Fig 4-3	AB AC BC	f1, f2, ..., f14	0.001	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	672
Fig 4-3	ABG ACG BCG	f1, f2, ..., f14	0.001	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	672
Fig 4-3	ABCG	f1, f2, ..., f14	0.001	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	224
Fig 4-3	AG BG CG	f1, f2, ..., f7	1000	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	336
Fig 4-3	AB AC BC	f1, f2, ..., f7	5	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	336
Fig 4-3	ABG ACG BCG	f1, f2, ..., f7	5	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	336
Fig 4-3	ABCG	f1, f2, ..., f7	5	0, 22.5, 45, 67.5, 90, 112.5, 135, 157.7, 180, 202.5, 225, 247.5, 270, 292.5, 315, 337.5	112
Total					3360

Classification of the Data Obtained from Faults for Training a Deep Neural Network in the Presence of Distributed Generation:

According to the table above, for each fault class at a single location, with sixteen angles and one fault resistance, the resulting matrix consists of 112 rows ($7 * 16 = 112$) and 2000 columns. This matrix includes seven rows representing three-phase voltage and current signals, along with one zero-sequence voltage signal. Consequently, for 14 locations and 16 angles with one fault resistance, the resulting matrix contains 1568 rows

and 2000 columns for a single fault type. Therefore, for ten fault classes, this results in a total of 15,680 rows and 2000 columns. Additionally, for the first seven locations, another fault resistance was tested, resulting in a matrix for each class, which, including the previous data, contains 2352 rows and 2000 columns. In total, the matrix provided as input to the neural network consists of 23,520 rows and 2000 columns.

The arrangement of each fault class in the data matrix is shown in Table 7:

This arrangement is the same for both cases; therefore, we will explain it only for the active power distribution system. As described in the data classification, each fault class consists of 2352 rows, and we will elucidate their priority order in the label character present in the structure created in MATLAB.

Table 7: The arrangement of each fault class in the data matrix.

Fault class number	1	2	3	4	5	6	7	8	9	10
Fault type	AG	BG	CG	AB	BC	AC	ABG	BCG	ACG	ABCG

After forming the data matrix, we also create a corresponding label for each row in each fault class, and then compile the data and labels into a structure as the primary dataset. In the following sections, we will see how the main data is called, and understanding the order and position of the data in this structure will aid us in comprehending subsequent stages more effectively.

When creating labels for the rows of the data matrix, it is important to note that, for instance, for the single-phase A to ground fault, we assign the label AG to each row related to this fault, regardless of whether it pertains to voltage or current. However, as previously mentioned, we know the specific signals included in every seven rows, and in subsequent stages, it is crucial to be familiar with their order for the integration of the extracted images. This is because each set of seven rows represents a distinct pattern for a type of fault at a particular angle and location, and any errors in the arrangement of signals and the integration of images could lead to unsatisfactory outcomes.

Stages of Creating RGB Images with Color Value JET128:

Initially, we retrieve the data matrix in the order described in previous sections using the corresponding code. Subsequently, the continuous wavelet transform is applied to each row of the matrix, that is, to each individual signal, followed by the application of the Hilbert transform. After this, each signal is converted into an RGB image with dimensions of 32 by 224 pixels, and these images are saved in a designated folder. Therefore, for each fault class in its respective folder, we now have 2,352 RGB images, each with dimensions of 224 by 32 pixels.

Assuming that the dimensions of each signal image are 224 by 224 pixels, it implies that after applying the wavelet transform and the Hilbert filter to each row of the data matrix—whose order we have specified—an image of dimensions 224 by 224 pixels, each with three channels, is generated.

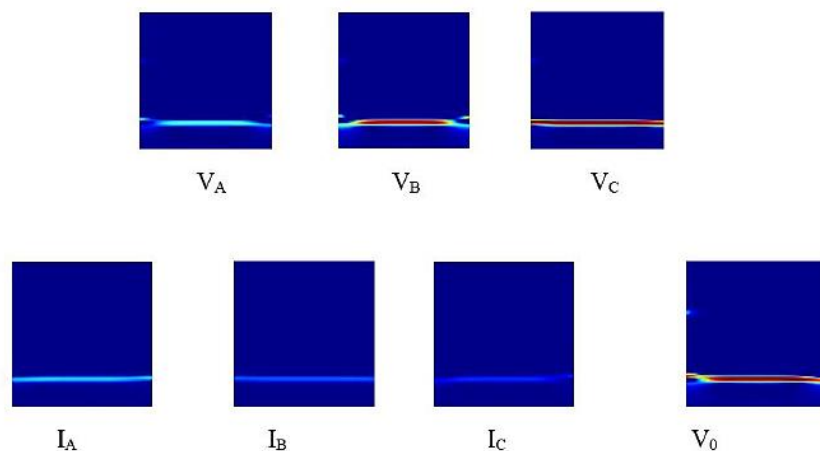


Fig. 13: Sample Images of single-phase fault signals A with dimensions of 224*224

The images in Fig. 13 show the single-phase A to ground fault

The images in Fig. 14 illustrates the single-phase B to ground fault

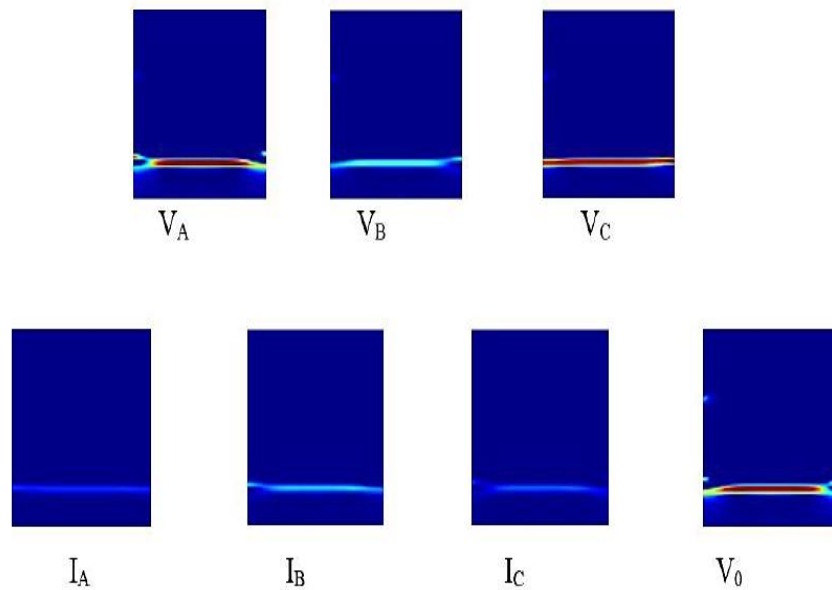


Fig. 14: Sample Images of single-phase fault signals B with dimensions of 224*224

Given the figures of both single-phase faults shown above and the similarity of some of them, if they are placed in a folder specific to their fault class, the neural network is likely to misclassify them, resulting in reduced accuracy. Initially, we attempted to address the issue in this manner, but the results were unsatisfactory. Therefore, the idea emerged that each class's images should have a specific order and arrangement.

Consequently, we created each signal image with dimensions of 32*224 pixels, arranging them such that the first three images represent the phase voltages A, B, and C, the next three images correspond to the phase currents A, B, and C, and the final image represents the zero-sequence voltage V_0 . This process allowed us to compile a square image of 224*224 pixels for each fault. In Fig. 15, we see the concatenated images of the aforementioned single-phase faults displayed together to form a single square image.

From the images, it is evident that the concatenation reduces the number of images from 2352 to 336 per class, which is seven times fewer. The result is a decrease in training time for the network, and the resultant images for each class become unique, facilitating easier recognition by the network and eliminating issues related to image similarity. In the following, we see the images of the remaining classes in Fig. 16:

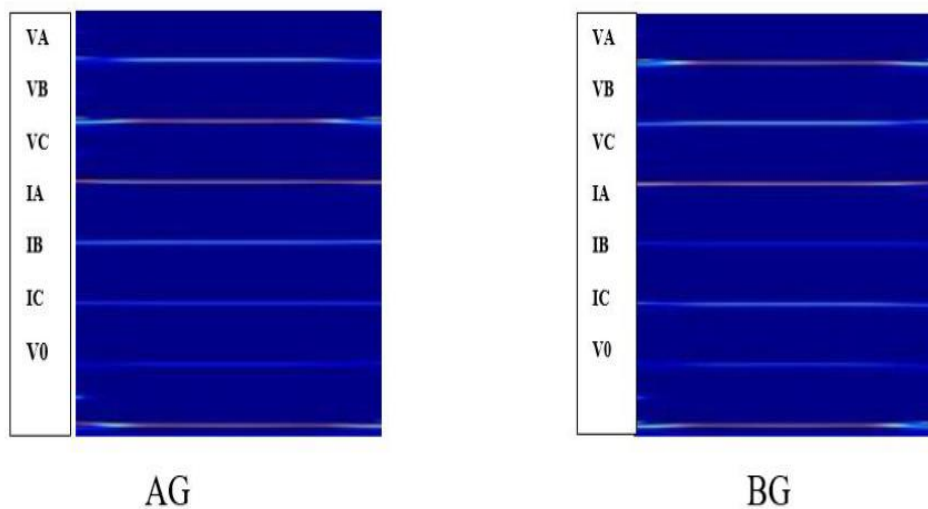


Fig. 15: An Example of Concatenated Images of Two Single-Phase Fault Signals

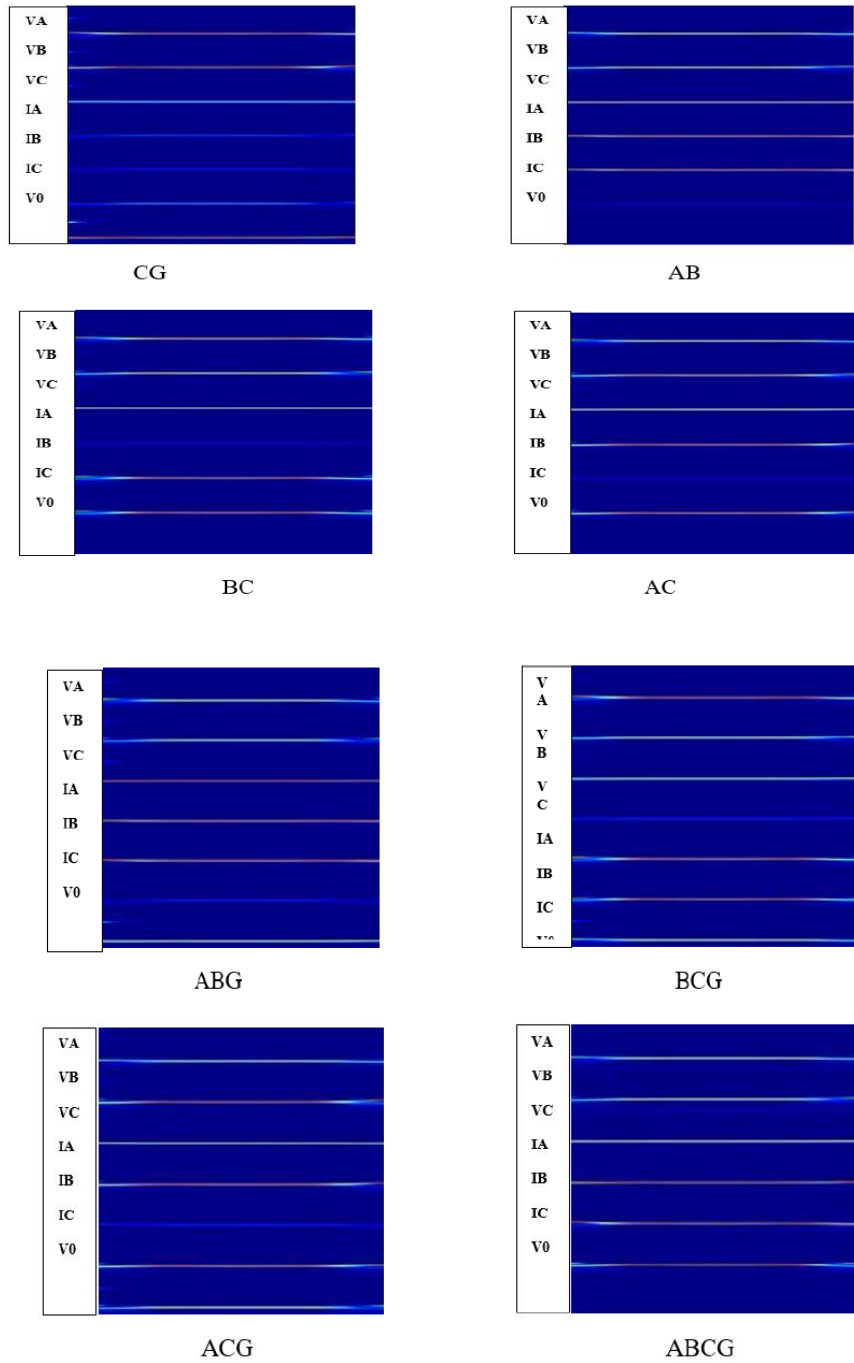


Fig. 16: Continuation of the images showing the remaining errors

4-3 Study Scenarios

Scenario 1 involves the detection of the type and location of faults in a passive power distribution system. Scenario 2 focuses on the same detection in a passive power distribution system with data noise at an SNR of 10 dB. Scenario 3 pertains to fault detection and localization in an active power distribution system. Finally, Scenario 4 examines fault detection and localization in the presence of distributed generation sources with data noise at an SNR of 40 dB.

Scenario 1

This scenario was studied at fourteen points, with three angles and one fault resistance, within a passive power distribution system. Following data collection using the methods described in previous sections, we obtained 42 images for each fault class. The results are illustrated in Figs. 17 and 18.

Fig. 17 shows the confusion matrix for Scenario 1, while Fig.18 illustrates the training process for this scenario.

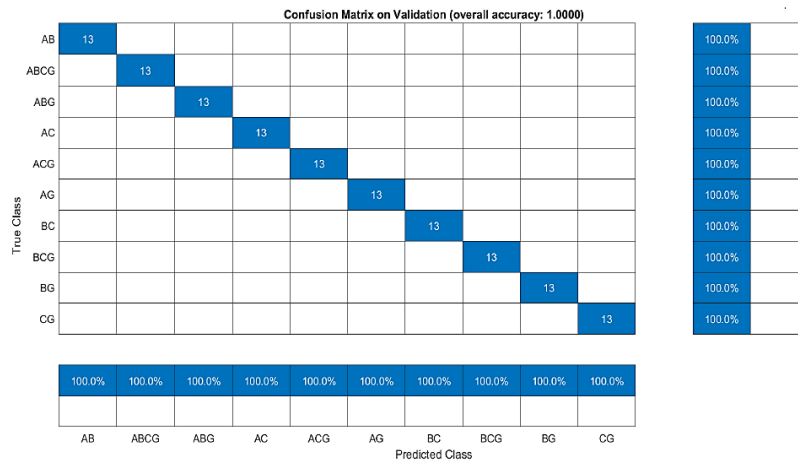


Fig. 17: Confusion Matrix for Scenario 1

As can be seen, the detection of the type and location of faults in this scenario was achieved with 100% accuracy.

The training process, illustrated in Fig. 18, shows that the training duration for this scenario was 1 minute and 12 seconds.

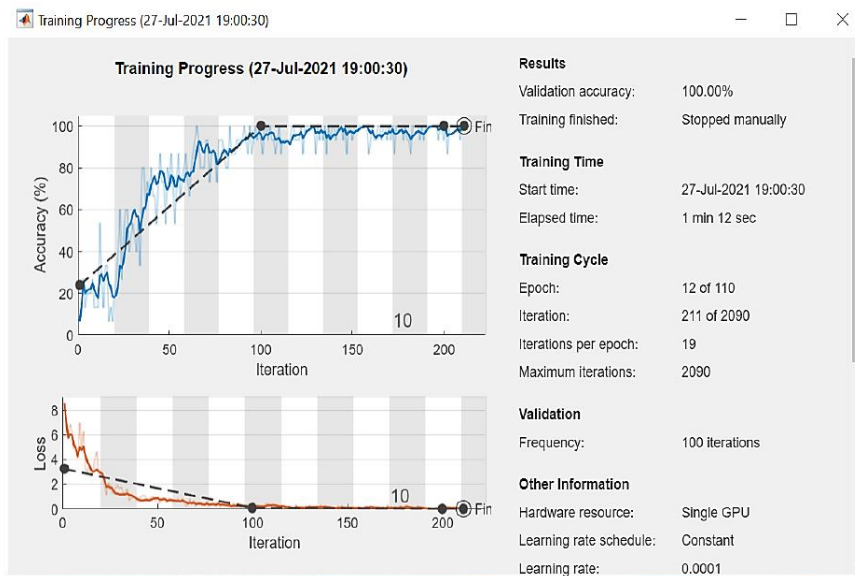


Fig. 18: Training Process for Scenario 1

Scenario 2

In this scenario, the detection of the type and location of faults in a passive power distribution system, with data noise at an SNR of 10 dB, was successfully achieved with 100% accuracy. Although the training duration increased to just one minute, it did not affect the accuracy of the network.

Fig. 19 presents the confusion matrix for this scenario:

In Figure 20, we observe the training process of the deep neural network in this scenario. As shown, the number of iterations is greater than that of Scenario 1, specifically 22 iterations were performed for the network to achieve 100% accuracy.

In both Scenario 1 and Scenario 2, a GPU was used for training.

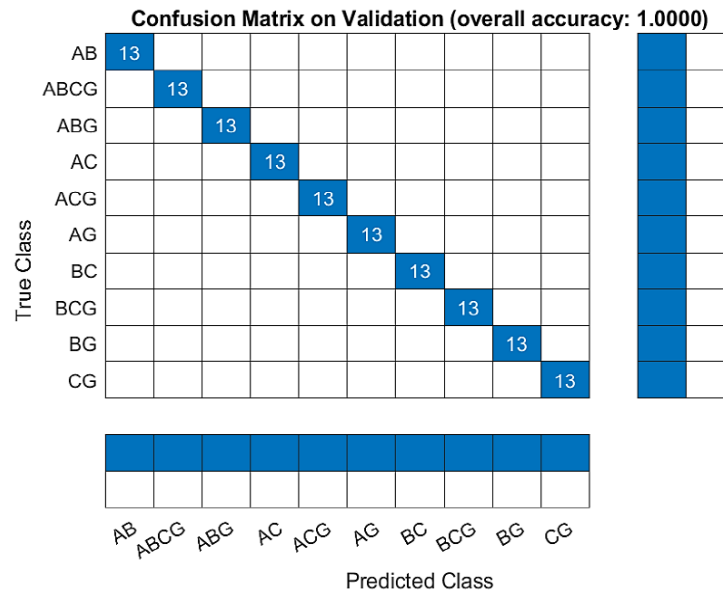


Fig. 19: Confusion Matrix for Scenario 2

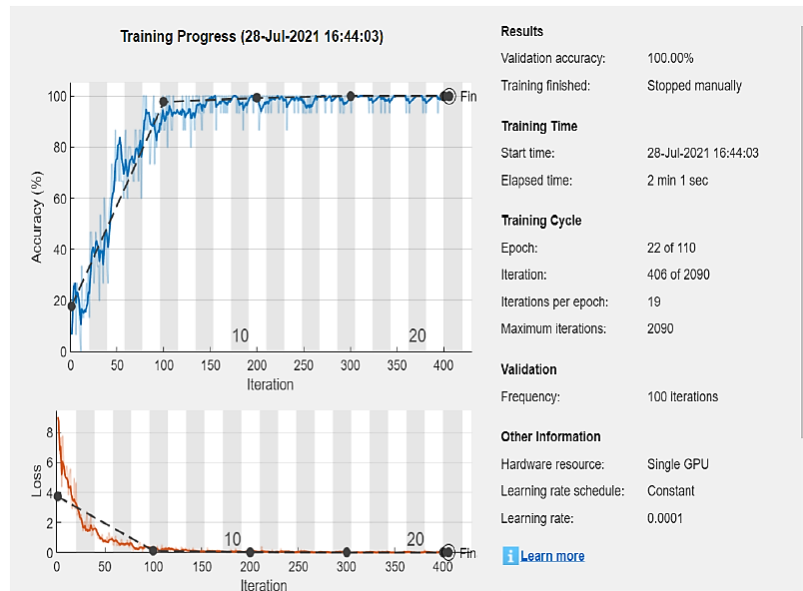


Fig. 20: Training Process for Scenario 2

Scenario 3

In Scenario 3, we performed fault detection and localization in an active power distribution system. The results obtained will be presented below. This scenario has two conditions:

In the first condition, the study was conducted at fourteen points, with sixteen angles and one fault resistance in the active power distribution system. After collecting data using the methods described in previous sections, we obtained 224 images for each fault class. The results for fault detection and classification are shown in Figs. 21 and 22.

Fig. 21 demonstrates the training Process of the Neural Network for the First Condition of Scenario 3, and Fig. 22 presents the confusion matrix for this condition.

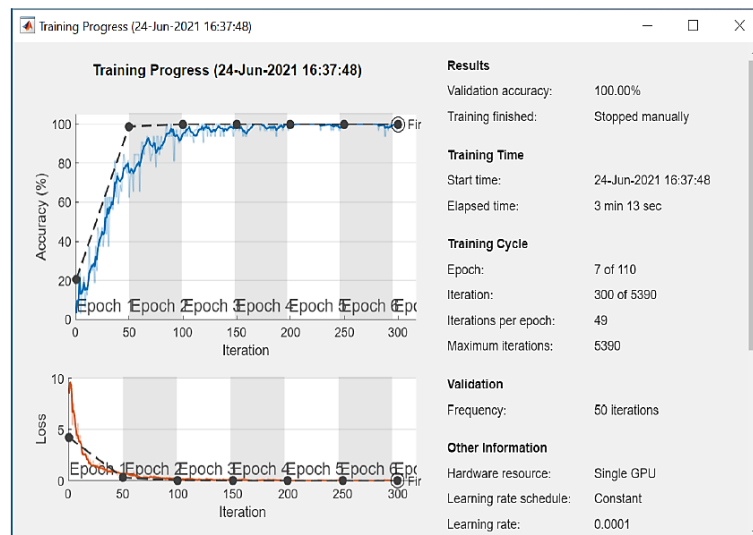


Fig. 22: Training Process of the Neural Network for the First Condition of Scenario 3

In the second condition of this scenario, the study was conducted at fourteen points, with sixteen angles and one fault resistance, as well as at seven of the initial points with a different fault resistance compared to the previous one. After collecting data using the methods described in previous sections, we have 336 images for each fault class. The results are presented in Figs. 23 and 24.

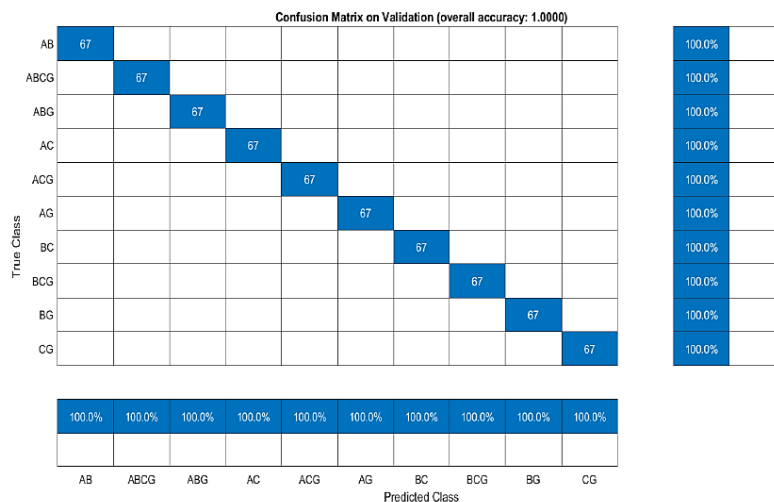


Fig. 22: Confusion Matrix for Validation of the First Condition of Scenario 3

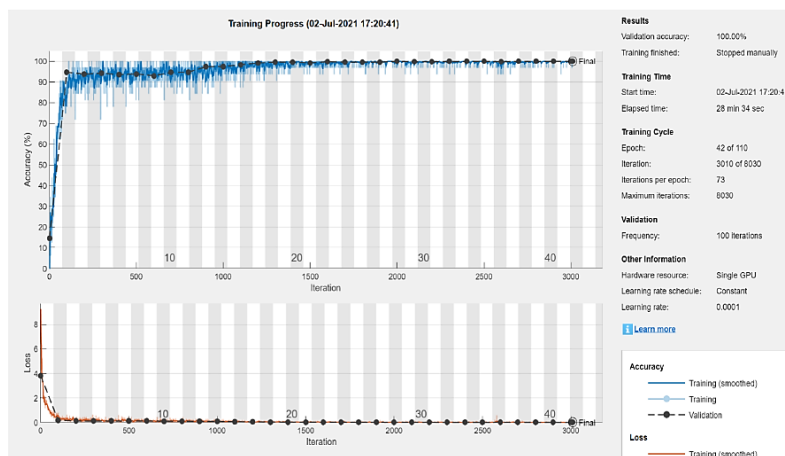


Fig. 23: Training Process of the Neural Network for the Second Condition of Scenario 3

The training process of the neural network for the second condition of Scenario 3 is illustrated in Fig.23.

The reason for the increased training time in this condition is the greater number and diversity of images compared to the first condition. The confusion matrix for the second condition of this scenario is presented in Fig. 24.

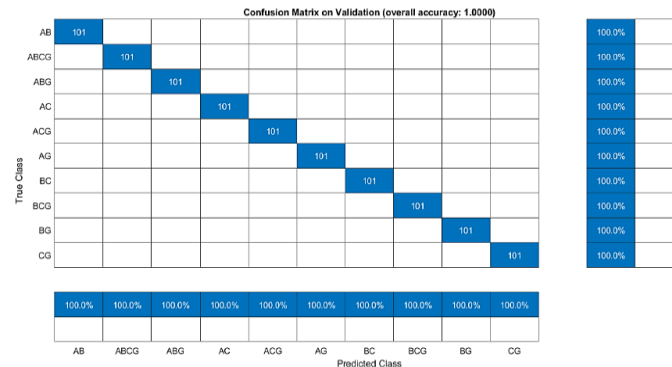


Fig. 24: Confusion Matrix for the Second Condition of Scenario 3

The test of the trained network for the second condition of Scenario 3 was conducted in locations where the network had not been trained. The results of this test can be found in Table 8.

Table 8: Test Results for the Second Condition of Scenario 3

Fault type	Number of feeder	Network prediction	Network accuracy	The distance of the fault occurrence from the beginning of the line
AG, BG, CG, AB, BC, AC ABG, BCG, ACG ABCG	1	TRUE	L1	2KM
AG, BG, CG, AB, BC, AC, ABG, BCG, ACG	1	TRUE	L1_1	3KM
ABCG	0.9997	TRUE	L1_1	3KM
ABCG, AG, BG,CG,AB,BC,AC	1	TRUE	L2	2.5KM
AG, BG, CG,BC,AC,BCG,ACG ABCG	1	TRUE	L3	5KM
AB	0.9993	TRUE	L3	5KM
ABG	0.9999	TRUE	L3	5KM
AG, BG, CG,BC, AC, ABG,BCG,ACG ABCG	1	TRUE	L4	6KM
AB	0.9999	TRUE	L4	6KM
AG, CG, AB, BC, AC ABG,BCG ABCG	1	TRUE	L5	0KM
BG	0.9997	TRUE	L5	0KM
ACG	0.9985	TRUE	L5	0KM
AG, BG, CG, AB, AC ABG,BCG,ACG ABCG	1	TRUE	L6	4KM
BC	0.9999	TRUE	L6	4KM

Localization for the Active Distribution Network on Feeder 2, which is 5 kilometers long, has been conducted. We will review the training and testing methodology together, considering each kilometer as a zone.

Fault points for neural network training

- F1: 0.25 km, 0.75 km, 1 km
- F2: 1.25 km, 1.75 km, 2 km
- F3: 2.25 km, 2.75 km, 3 km
- F4: 3.25 km, 3.75 km, 4 km
- F5: 4.25 km, 4.75 km, 5 km

Faults were categorized into 10 classes, with each class representing 16 angles at each location. In total, we have 5 spatial classes, each containing 480 images of various types of faults related to that section of the feeder. The neural network used for localization is the same one employed for fault detection and classification, with the distinction that its final layer, which is fully connected, consists of 5 classes.

Fig. 25 presents the confusion matrix for localization on Feeder 2 in Scenario 3, demonstrating an accuracy of 100%.

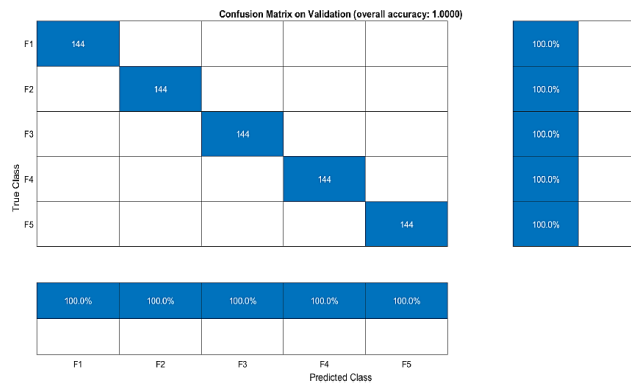


Fig. 25: Confusion Matrix for Localization in Scenario 3

Fig. 26 illustrates the localization training process, which took 42 minutes and 23 seconds for the 5 spatial classes.

Table 9 shows the test results of the trained network in intervals where it had not been trained. Notably, all fault occurred at a single angle.

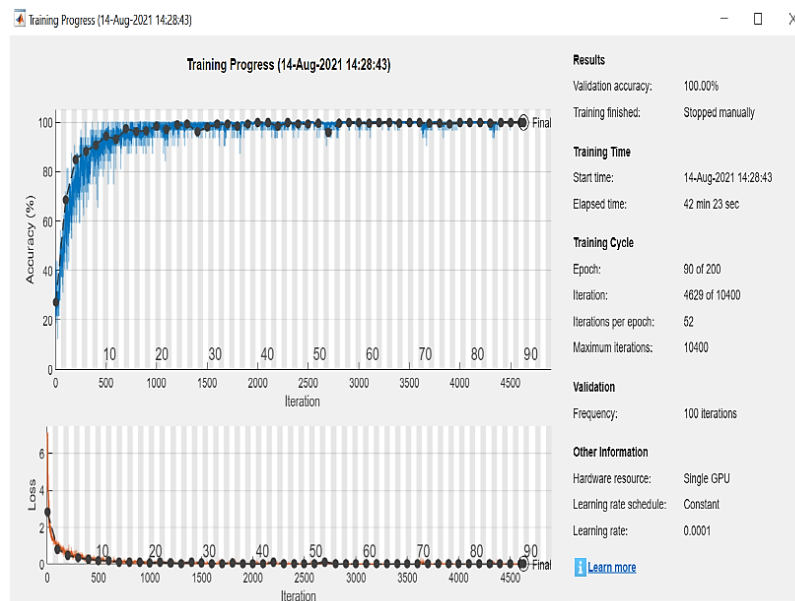


Fig. 26: Localization Training Process for Scenario 3

Table 9: Results of the Localization Test

Fault type	Accuracy	Network prediction	The distance of the fault occurrence from the beginning of the line
AG,BG,CG AB,BC,AC ABG,BCG,ACG ABCG	1	F1	0.5KM
ABCG	0.99	F2	1.5KM
AG,BG,CG AB,BC,AC ABG,BCG,ACG	1	F2	1.5KM
AG,BG,CG AB,BC,AC ABG,BCG,ACG	1	F3	2.5KM
ABCG	99.99	F3	2.5KM
ABCG	99.97	F4	3.5KM
AG,BG,CG AB AC	1	F4	3.5KM
BC	0.9999	F4	3.5KM
ABG,BCG,ACG	1	F4	3.5KM
AG,BG,CG AB,BC,AC ABG,BCG,ACG	1	F5	4.5KM
AB,BC,AC ABG,BCG,ACG	1	F5	4.5KM
ABCG	0.9985	F5	4.5KM

Scenario 4

In Scenario 4, the detection of the type and location of faults in the presence of distributed generation sources was performed, with data noise at an SNR of 40 dB. The results are presented alongside those from previous scenarios. Given the level of noise applied to the data, the results are acceptable.

The confusion matrix for this scenario is shown in Fig. 27:

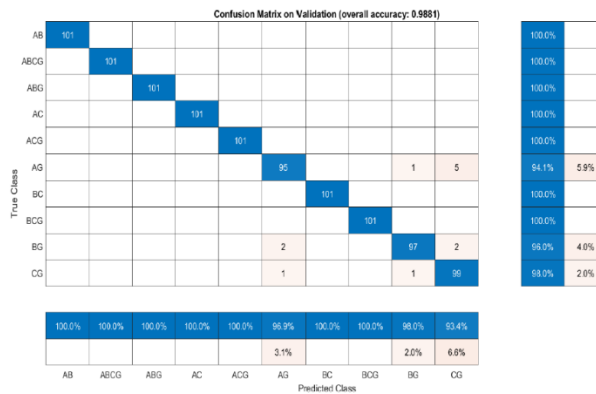


Fig.27: Confusion Matrix for Scenario 4

The training process of the neural network, considering the level of noise applied, has taken more time to achieve the desired results. Figure 28 effectively illustrates this process. Furthermore, a GPU was utilized for this process across all scenarios. As shown, the overall accuracy of the network has reached 98.81%, and the increased noise has impacted the prediction accuracy for the three classes of single-phase faults to ground.

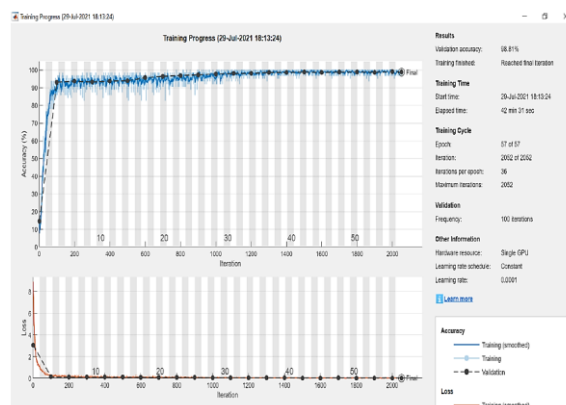


Fig. 28: Training Process of the Neural Network for Scenario

5- CONCLUSION

5-1 Summary of Scenario Results

As mentioned in the previous chapter, this paper successfully addresses some of the most challenging scenarios with high accuracy using a deep neural network (CNN) specifically designed for this problem. Thus, easier scenarios have not been discussed, as it is evident that the method presented in this paper yields satisfactory results much more easily for simpler problems.

- In Scenario 1, for fourteen specified locations on the studied distribution system, various types of faults were simulated at three angles and with a fault resistance. After feature extraction and converting the signals into RGB images, a composite image of 224 by 224 pixels was created for training the deep neural network by merging the seven signals considered for each fault. These seven signals include three-phase voltage and current signals, as well as zero-sequence voltage signals. As observed in the training process of the deep neural network, the training time in this scenario was significantly shorter compared to Scenario 2, due to the higher quality of the images. Consequently, the deep neural network reached 100% accuracy more quickly.
- In Scenario 2, the issue of fault type and location detection was addressed by adding noise to the images at an SNR of 10 dB. Due to the introduction of noise, the training process time and the number of training iterations required to achieve 100% accuracy increased.
- In Scenario 3, two conditions exist. In the first condition, the study was conducted at fourteen points, with sixteen angles and one fault resistance in the active power distribution system. After collecting data using the methods described in previous sections, there were 224 images for each fault class, achieving fault type and location detection with 100% accuracy. Additionally, localization in this condition, which involved the presence of distributed generation sources, was successfully conducted with 100% training accuracy across five zones on Feeder 2. The trained network was then successfully tested in locations not included in the training data. In the second condition of this scenario, faults were created at fourteen specified locations on the studied distribution system in sixteen angles and with one fault resistance. Additionally, this was also done in seven of the initial locations with a different fault resistance compared to the previous one, to enhance the training of the deep neural network regarding different fault types. In this case, there are 336 images for each fault class; however, achieving 100% accuracy in training the deep neural network required more iterations and extended training time.
- In Scenario 4, the detection of fault type and location in the presence of distributed generation sources was performed with data noise at an SNR of 40 dB. Considering the level of noise applied to the data, the result of 98.81% is acceptable. The applied noise primarily impacted single-phase to ground faults.

5-2 Conclusions from the Analysis of the Presented Scenarios

1. Given the increasing complexity of power distribution systems and the ineffectiveness of traditional methods, these scenarios have demonstrated that even the most challenging issues can be resolved using artificial intelligence. These methods could contribute to the stability of power delivery to consumers and prevent damage to both consumers and distribution companies.
2. We no longer have concerns regarding the detection and localization of faults due to the presence of distributed generation sources in the distribution system. This allows us to safely leverage the benefits of these sources, which are increasingly becoming part of the network.
3. To address challenging and nonlinear problems, such as the paper issue, there is no need to solve complex equations and difficult mathematical problems. It is important to mention that feature extraction plays a crucial role in solving these problems using deep neural networks.
4. There is no longer a need for many of the hardware and devices currently used for this issue.

Utilizing a high-performance processing system allows for the easy resolution of these types of problems.

5. By measuring voltages and currents at the main substation, there is no longer a necessity to employ multiple measuring devices within the network.
6. Although we also attempted to solve this problem using a Deep Belief Network (DBN) and obtained a somewhat reasonable result, it was not acceptable. While a detailed discussion of this in the paper is unnecessary, it is important to note that the experience gained here is incomparable to that of a Convolutional Neural Network (CNN), which can be confidently stated as superior to all other deep neural networks.

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